Artificial Effort

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Abstract

Real-effort tasks, in which participants perform cognitively costly activities whose outcomes depend on actual performance, are widely used in experimental economics. Their validity, however, rests on the assumption that a human performs them. We study whether this assumption still holds in the era of Artificial Intelligence (AI) and Large Language Models (LLMs). Using 8 canonical real-effort tasks and 23 LLMs from three major providers, we show that most tasks can now be solved accurately and at a negligible cost, while only a few resist automation. Within the set of models we test, newer releases generally outperform older ones, and mid-tier models perform close to frontier ones, broadening the set of widely accessible models that can automate these tasks. Additionally, we show that verbally offering monetary incentives has no effect on LLM performance. Our findings establish a boundary condition for the use of real-effort tasks in unsupervised settings: when participants can cheaply outsource task completion to an LLM, observed performance may no longer reflect genuine human effort.

Keywords: Large Language Models, Real-effort tasks, Experiment

JEL Codes: C81, C88, C90, D90.

1 Introduction

Real-effort tasks are a central methodological tool in experimental economics (Charness et al., 2018; Carpenter and Huet-Vaughn, 2019). By requiring participants to complete activities that take time, attention, and cognitive processing, these tasks are used to study effort provision, performance, fatigue, and responsiveness to incentives. Their appeal lies in the fact that outcomes are not purely stated: performance is intended to reflect the costly exertion of resources. For this reason, real-effort tasks are often treated as a relatively direct way to measure behavior-under-effort costs rather than preferences reported in the abstract. This interpretation, however, rests on an important and often implicit assumption: *the agent completing the task should be a human.*

The rapid diffusion of modern Artificial Intelligence (AI) systems and, in particular, of Large Language Models (LLMs) such as ChatGPT (OpenAI, 2022), Gemini (Pichai and Hassabis, 2023; Team et al., 2023), or Claude (Anthropic, 2023) makes this assumption less secure, especially in online environments. Indeed, in remote and unsupervised settings such as online labor markets, open survey links, and web-based experiments, researchers may be unable to verify with confidence whether a response was generated by a human participant or by an automated system (Bevendorff et al., 2025). If LLMs were able to complete canonical real-effort tasks with high accuracy at low marginal cost, then these tasks may cease to function as measures of costly human effort in precisely those contexts where they are often most attractive to use, and could also be exploited by malicious individuals as a source of revenue in a *task–milling* action. The concern is therefore not only one of participant authenticity but, rather, a question of construct validity: a task designed to capture human effort may instead capture access to a capable AI model.

This concern has become more pressing as online experimentation has expanded more rapidly than the tools available to verify participant identity and ensure proper task completion. Yet, there is still little direct evidence in the literature that state-of-the-art LLMs can effectively solve those real-effort tasks that are widely used in experimental economics, with a notable exception of how AI can replace humans in generating survey responses (Celebi et al., 2026).[1]

This paper provides an assessment of whether LLMs can perform well on the canonical tasks that experimental economists routinely adopt to measure human effort. We evaluate a set of well established real-effort tasks using three families of LLMs taken from the GPT, Gemini, and Claude series and address four related research questions. First, we compare accuracy across LLMs to assess whether their performance differs and, if so, how substantially. Second, we examine how performance evolves across different versions of the same model family, which is an indicator of how quickly accuracy may change over time as LLMs improve. Third, we consider the economic dimension of substitution by comparing model performance to the cost of model use, asking whether

[1] On the other hand, success on standard AI benchmarks—whether they test general knowledge and reasoning (e.g., MMLU (Hendrycks et al., 2021a), BIG-bench (Srivastava et al., 2022), ARC (Clark et al., 2018), HellaSwag (Zellers et al., 2019)), mathematical logic (e.g., GSM8K (Cobbe et al., 2021) MATH (Hendrycks et al., 2021b)), factual retrieval (e.g., TriviaQA (Joshi et al., 2017) and Natural Questions (Kwiatkowski et al., 2019)), or coding and agentic capabilities (e.g., HumanEval (Chen et al., 2021) and SWE-bench (Jimenez et al., 2024))—does not automatically imply success on the specific tasks economists use to elicit costly effort.

replacing human effort with LLM-generated output is not only feasible but also cost-effective for a profit-seeking agent. Fourth, we test whether the verbal promise of a monetary reward associated with task performance may affect LLM accuracy—a question central to the logic of real-effort experiments, which rely on incentive sensitivity.

Our results are as follows. First, we show that LLM performance varies substantially across models, determined largely by model tier rather than provider. Second, we show that, within each LLM tier (e.g., different versions of Claude Sonnet), with only one exception newer releases always outperform previous ones, with the largest gains concentrated in the mid-range tiers. Third, we show that, at typical online-experiment piece rates, every model costs only a small fraction of what a human participant would earn for the same workload, thereby generating a positive net gain. Lastly, we show that verbal monetary incentives have no detectable effect on LLM performance, and neither do instructions to behave as a human participant.

2 Method

Our benchmark comprises eight real-effort tasks that are available in the Otree repository (Chen et al., 2016), implemented in their canonical form or with only minor modifications (Charness et al., 2018). Each task requires a different combination of perceptual, arithmetic, and reasoning abilities, as summarized in Table 1.[2]

Table 1: Summary of the eight real-effort tasks.

Task	**Input**	**Problem**
Addition (Niederle and Vesterlund, 2007)	An addition expression with 3 integer addends of up to 3 digits each.	Calculate the exact sum.
Counting Zeros (Abeler et al., 2011)	A 15×10 table of 0s and 1s.	Count the number of 0s.
Letter Decoding (Erkal et al. 2011;)	A table mapping 10 letters to the digits 0–9, together with a 7-digit coded string.	Decode the string back into the corresponding letters.
Number Pair Summation (Rosaz et al., 2016)	A 4×4 grid of decimal numbers.	Identify and return the one pair that sums to 10.0.
Sequence Completion (Bracha and Fershtman, 2013)	A sequence of numbers following a mathematical pattern.	Predict the next element.
Sudoku (Calsamiglia et al., 2013)	A partially filled 6×6 Sudoku grid (with 2×3 boxes).	Identify all missing numbers and report them in reading order, space-separated.
Distorted Text (Augenblick et al., 2015)	An image of a 12-character alphanumeric string rendered with random distortions.	Transcribe the text exactly (case-insensitive).
String Entry (Kessler and Norton, 2016)	A 9-character string of special characters (`/ \ ) ( _ <` and space) displayed as an image.	Reproduce the exact string, including spaces.

[2]The pre-registration plan is available on OSF at https://osf.io/7pz4m/overview. Code and data for replication are available at https://github.com/belerico/llm-real-effort; the project website summarizing the main results is available at https://belerico.github.io/artificial-effort/.

2.1 Tested Models and Parameters

Our experiments evaluate 23 state-of-the-art Large Language Models (LLMs) from three major providers—OpenAI, Google, and Anthropic—spanning a range of capabilities from lightweight to frontier reasoning models. All selected models support multimodal (text and image) inputs and visual reasoning. To ensure a standardized interface, all models are accessed via the OpenRouter API.[3] The full list of models, release dates, and pricing details are reported in Appendix C.1 (Table 5).

LLMs process inputs in tokens—sub-word text units or visual patches—whose granularity varies across providers.[4] Pricing depends on the number of input and output tokens. For reasoning-capable models (LLMs that produce an internal chain of thought before outputting their answer), intermediate reasoning tokens (i.e., the internal chains of thought, which is not shown in final output) may also be generated and are billed at output-token rates.

To ensure comparability, we impose uniform parameters across models: we set a maximum of 2,048 output tokens (including reasoning tokens) and a time limit of 120 seconds; responses exceeding either constraint are treated as incorrect. For models with adaptive reasoning budgets, we set the reasoning effort to "high", allowing up to 80% of the token limit to be allocated to reasoning. All requests are issued through the OpenRouter Chat Completions API. We set the sampling temperature to 0.6 for models that expose it (Anthropic and Google).[5] We do not set top-p, which OpenRouter defaults to 1.0, and we do not set a random seed. Each of the 20 trials of a model–task pair is an independent, stateless API call carrying no history from previous trials; with a non-zero temperature, trials are therefore independent and stochastic draws rather than deterministic copies. We note, however, that providers may update the snapshot behind a model identifier over time, so strict identical-distribution across long time spans is not guaranteed.

2.2 Experimental Setup

In what follows, we use the following terminology. A *task* is one of the eight real-effort task types listed in Table 1 (e.g., *Addition*). A *trial* is one freshly generated instance of a task, completed once by one model. For every model–task pair we perform 20 trials. Unless stated otherwise, all accuracy, cost, and net-gain quantities are reported *per task*, i.e., for the full 20-trial model–task pair rather than for a single trial.

Our pipeline works as follows. For each trial of a task, the oTree back-end generates the task instance and renders it as a standard HTML page. We grab the rendered input (textual instructions and the image representing the task instance) and send it to the LLM through the OpenRouter API. The LLM's textual response is parsed and passed back to the oTree back-end for correctness verification.[6]

[3] `https://openrouter.ai`

[4] `https://help.openai.com/en/articles/4936856-what-are-tokens-and-how-to-count-them` and `https://ai.google.dev/gemini-api/docs/tokens`

[5] OpenAI reasoning models do not support a temperature control, so the parameter is omitted and OpenRouter applies a default of 1.0.

[6] This workflow mirrors how a human participant could use an LLM in practice: screenshotting the task page and copying the LLM's answer into the oTree input field. Fully automating this interaction—e.g., via browser-automation tools such as Playwright (`https://playwright.dev`) is feasible and would

~~For each LLM, we repeat each task for 20 trials.~~ An answer is scored as correct only if it exactly matches the ground-truth solution; no partial credit is awarded. From the fraction of correct answers, we compute the accuracy of each model to solve each task. We also collect the the total number of tokens used to elaborate the answer, as well as quantify the cost of performing each task.

2.3 Treatments

As a further objective, we check whether the LLMs we test are sensitive to the verbal offer of monetary incentives, and whether their behavior aligns with that of human participants (despite the absence of actual rewards). To this end, we employ a 2×2 design that crosses the specification of monetary pay for each correct answer with the request to simulate human behavior. We implement these four treatment conditions (T1 to T4) alongside our main control setting (T0), where we specifically ask the LLM to solve the task using a neutral prompt, omitting any framing regarding human behavior or monetary incentives. The exact prompts we used are reported in Appendix A.1.

3 Results

Throughout this section, each model–task pair comprises 20 independent trials. Unless otherwise noted, the following figures report LLM performance under the control treatment (T0), which uses a minimal prompt devoid of any incentive or human framing, providing the cleanest measure of each LLM's capability.

3.1 Result 1: Task difficulty

First, we notice that not all real-effort tasks are equally challenging for LLMs, and that some cognitive demands remain beyond current LLMs' capabilities.

In Figure 1, we report the average accuracy across all 23 LLMs under T0. Four tasks are near-ceiling: *Addition* (100%), *Number Pair Summation* (97.6%), *Letter Decoding* (94.1%), and *Sequence Completion* (90.0%). These tasks involve small, well-structured inputs that can be solved through a short sequence of elementary operations: adding three small numbers, searching over 120 candidate pairs, applying a lookup table, or recognizing a pattern in a short sequence. Prior work published within the AI literature has shown that modern LLMs can reliably solve these kinds of problems (Nogueira et al., 2021; Wei et al., 2022).

At the other extreme, *Sudoku* (25.0%), *Counting Zeros* (27.2%), and *String Entry* (37.0%) prove substantially harder. The three tasks that remain challenging expose limitations that go beyond visual perception. *Sudoku* requires multi-step reasoning across rows, columns, and boxes simultaneously—it's a combinatorial problem that

add negligible latency and no additional token cost. A related approach is taken by Celebi et al. (2026), who deploy full browser-automation agents that control mouse movements, keystrokes, and page navigation to complete online surveys. While their work demonstrates that LLM-driven agents can mimic realistic human interaction patterns, the reliance on agent-mode orchestration introduces substantial overhead in both cost and latency, limiting its scalability and its economical viability. Our pipeline bypasses the browser and communicates with the LLM via API at a fraction of the cost. See Appendix B for a reference implementation.

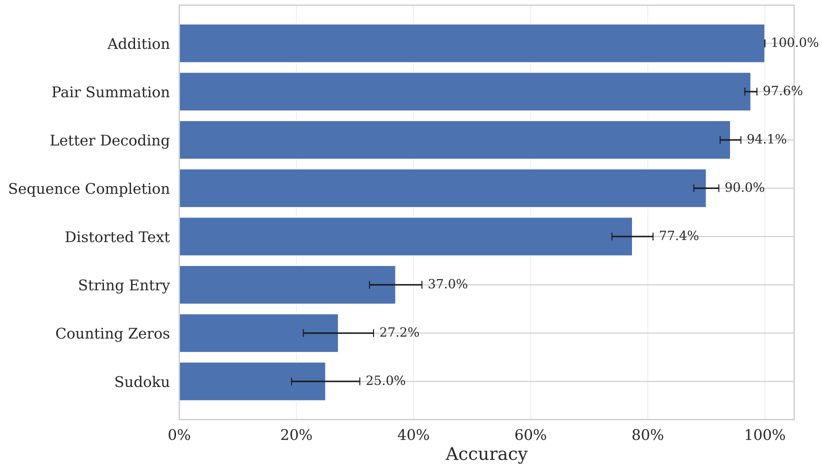

Figure 1: Average accuracy per task under the control treatment (T0), pooled across all 23 LLMs. Error bars denote standard errors.

is inherently hard for LLMs regardless of input format (Maiya et al., 2025), though vision/parsing errors from misreading the board may also contribute. *Counting Zeros* demands tallying individual cells in a large grid, a task at which LLMs struggle due to known architectural limitations on counting and enumeration (Yehudai et al., 2024; Fu et al., 2024). *String Entry* requires exact reproduction of special characters and spaces, which LLMs tend to silently alter or reformat, producing near-matches that score poorly.

Distorted Text Transcription falls in the intermediate range (77.4%), consistently with its mixed demands—character recognition is largely tractable (it is, indeed, the most-famous practical problem to be successfully solved by an early neural network—see LeCun et al. (2002)), but random distortions introduce perceptual noise that degrades accuracy.[7] This difference in difficulty has a clear methodological implication: tasks that the behavioral economics literature has, traditionally, considered cognitive demanding (e.g., Addition) are also the ones most vulnerable to LLM automation, while less-demanding tasks requiring visual search (e.g., *Counting Zeros*) happen to exploit a modality where current LLMs are weak.

3.2 Result 2: Performance across providers, tiers, and tasks

Figure 2 reports the average accuracy across the eight real-effort tasks for all 23 LLMs we tested under the control treatment (T0), with models grouped by provider and tier; darker shades indicate newer releases within the same tier. Figure 3 reports the accuracy under T0 for every model–task pair. From the figures, three main patterns emerge.

First, while the best top-tier models—Claude Opus 4.5 and 4.6, Gemini 3.1 Pro, and GPT-5.4—achieve average accuracies above 80%, mid-tier ones (Claude Sonnet, Gemini Flash, GPT Mini) are not as strong but still achieve a good performance: the difference,

[7]Our results align with recent evidence that multimodal LLMs handle text and object-recognition CAPTCHAs reasonably well, but struggle with those requiring image-based, spatial or multi-step reasoning (Ding et al., 2025; Wang et al., 2025; Zhang et al., 2025).

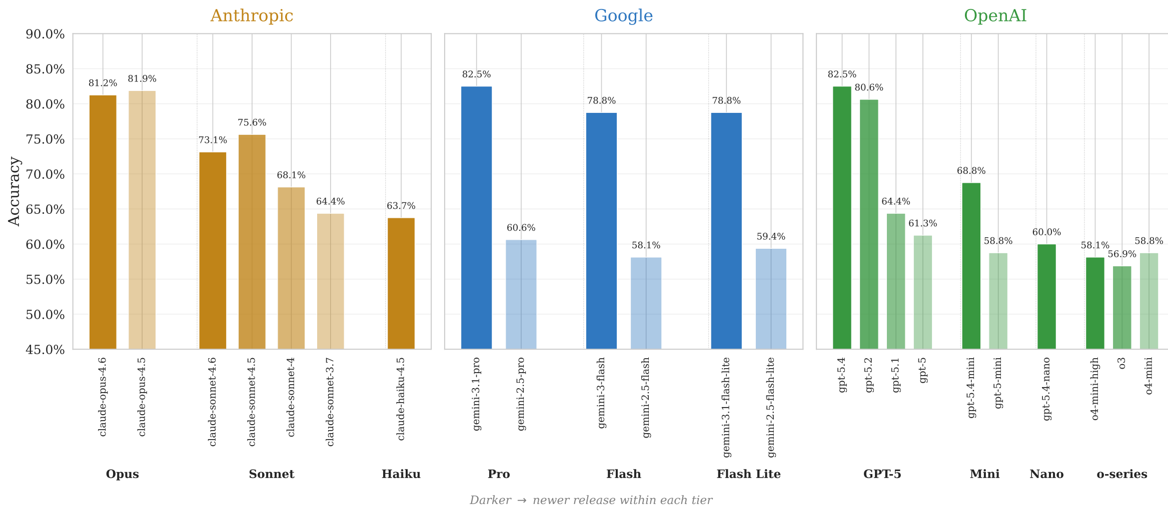

Figure 2: Average accuracy by LLM within each provider family under the control treatment (T0), ordered chronologically by release date within each models tier. Each bar represents a single model's accuracy averaged over all tasks.

per provider, between a best-performing mid-tier LLM and a best-performing top-tier one varies between 4% and 14%. This means that the set of models that perform well on real-effort tasks is expanding from the more-expensive top-tier models to cheaper and more widely accessible ones.

Second (and as one may except), later versions of a given tier generally have higher performance.[8] This suggests that, as providers continue to release stronger models, tasks that are currently resistant to automation may not remain so.

Third, as shown in Figure 3 and reflecting the analysis in Section 3.1, an LLM averaging 80% overall may score near-perfectly on some tasks (e.g., *Addition* and *Pair Summation*), while failing on others (e.g., *Sudoku* and *Counting Zeros*). This indicates that some tasks are currently resistant to automation, but also suggests that the set of tasks that are not will expand with new model releases.

3.3 Result 3: Cost–accuracy trade-offs and the economics of substitution

Beyond accuracy, the practical viability of LLM-based automation depends on the cost of running the LLMs incurred by the profit-seeking agent.

Figure 4 visualizes the trade-off between the average LLM cost per task and the total number of tokens consumed. We define the total cost for model m on task t as $C_{m,t} := \sum_{i=1}^{N} \left(p_m^{\text{in}} \cdot T_{m,t,i}^{\text{in}} + p_m^{\text{out}} \cdot T_{m,t,i}^{\text{out}}\right)$, where N is the number of trials, $T_{m,t,i}^{\text{in}}$ and $T_{m,t,i}^{\text{out}}$ are the input (textual prompt plus the image representing the particular task instance) and output (reasoning plus answer) tokens for model m on the i-th trial of task t, and p_m^{in}, p_m^{out} are the per-token input and output prices for model m. The average cost per model is then $\bar{C}_m := \frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} C_{m,t}$, where $\mathcal{T}$ is the set of tasks.

[8] Gemini Pro improved by 22 percentage points between versions 2.5 and 3.1, GPT-5 by 21 percentage points from GPT-5 to GPT-5.4, and Claude Sonnet by 11 percentage points from version 3.7 to 4.5 (though version 4.6 shows a slight regression).

Figure 3: Accuracy (%) per LLM and task under the control treatment (T0). LLMs are grouped by provider family (Anthropic, Google, OpenAI) and ordered by accuracy within each family (best at the top); tasks are ordered by difficulty (easiest on the left). Black lines separate provider families.

The cost to achieve the (current) best performance, as shown in Figure 4, is modest in absolute terms: GPT-5.4 and Gemini 3.1 Pro achieve the highest accuracy over all 20 trials (82.5%) at an average cost $\bar{C}_m$ less than \$0.25—a fraction of what a human participant would earn on a typical experiment in an platforms like Prolific or MTurk for an equivalent workload. At the same time, the cheapest LLMs still achieve an accuracy of 58–60%, meaning that even the most budget-constrained automation attempt would produce correct answers on the majority of tasks. Notably, Gemini 3.1 Flash Lite Preview achieves 78% accuracy at an average cost $\bar{C}_m$ of \$0.03, suggesting that the cost barrier to high-quality automation is already negligible.

3.3.1 Optimizing performance

Notably, the assumption, made in Figure 1, of a single LLM being used for all tasks may lead to an *underestimation* of the threat of automation. This is because a strategic participant could select different LLMs for different tasks—using, for example, Gemini 3.1 Pro for *Counting Zeros* (85%), GPT-5.2 for *Sudoku* (85%), and Gemini 3.1 Pro for *String Entry* (75%). By cherry-picking the best-performing model for each task, the participant would achieve an average accuracy of 93.1%—an 11% improvement over the choice of a single model (Gemini 3.1 Pro and GPT-5.4, 82.5%). Under this "optimal participant" scenario, even the hardest tasks in our benchmark (*Sudoku*, *Counting Zeros*, and *String Entry*), which Section 3.1 identified as resistant to individual LLMs, are solved at 75% accuracy or above under cherry-picking. Therefore, the automa-

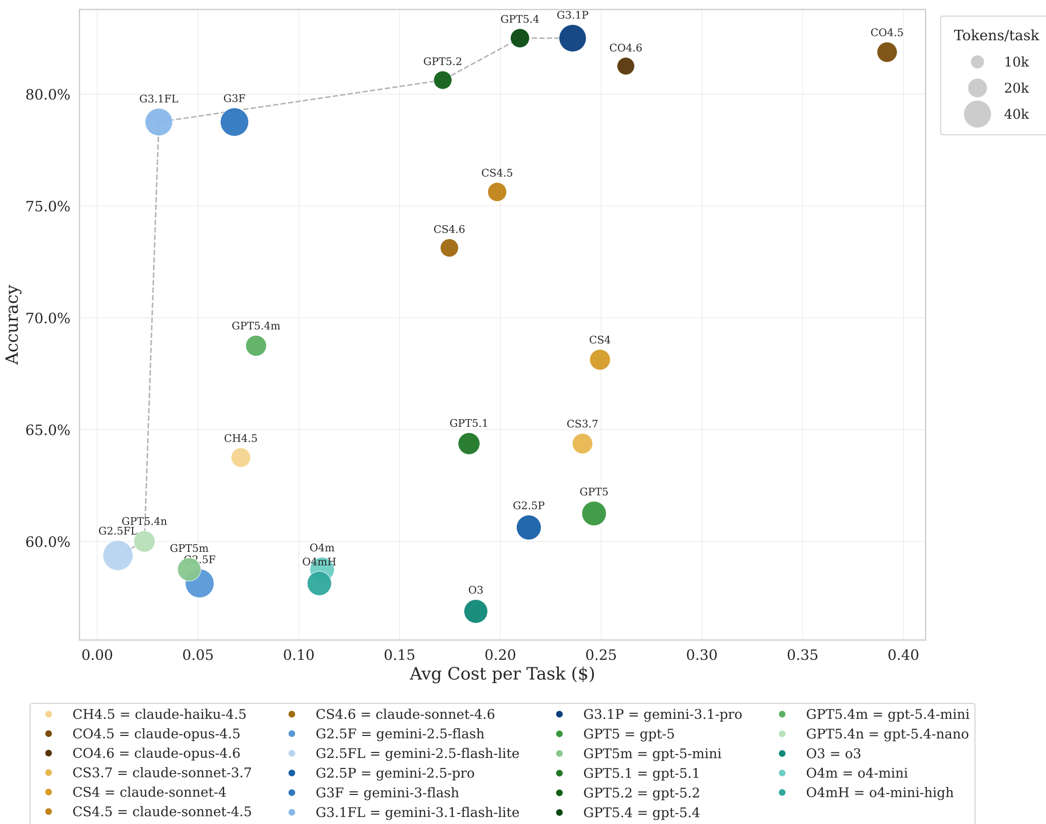

Figure 4: Cost–accuracy–tokens trade-off under the control treatment (T0). Each bubble represents a model; horizontal position indicates the average cost per task $\bar{C}_m$ (\$), vertical position indicates average accuracy, and bubble size is proportional to the total number of tokens consumed (prompt plus output) across 20 trials per task.

tion resistance of a task strongly depends on the will of a knowledgeable profit-seeking participant to use the right LLM to solve it.

Figure 5 formalizes this point by computing the *net gain per task*, defined as the hypothetical payout (\$0.25 per correct answer, a conservative estimate of typical online-experiment piece rates) minus LLM cost (reported in Figure 12). Crucially, *every* model we tested generates a positive net value, ranging from \$2.66 (o3) to \$3.92 (GPT-5.4). Such a universally positive net gain confirms that, at compensation rates typical of online experiments, replacing human participants with LLMs is not only feasible but economically dominant.[9]

3.4 Result 4: Treatment differences

In Appendix A.2, we report the effects of prompting LLMs by verbally offering different monetary incentives and/or explicitly instructing them to behave like human partici-

[9]As already mentioned, our evaluation assesses the LLM's ability to solve tasks from pre-captured screenshots and text. In practice, fully-automated pipeline would be even simpler and would consume even less tokens due to the fact that many tasks could be encoded by a purely-textual prompt. This implies that the economic advantage documented in Figure 5 is a conservative estimate of the true substitution incentive.

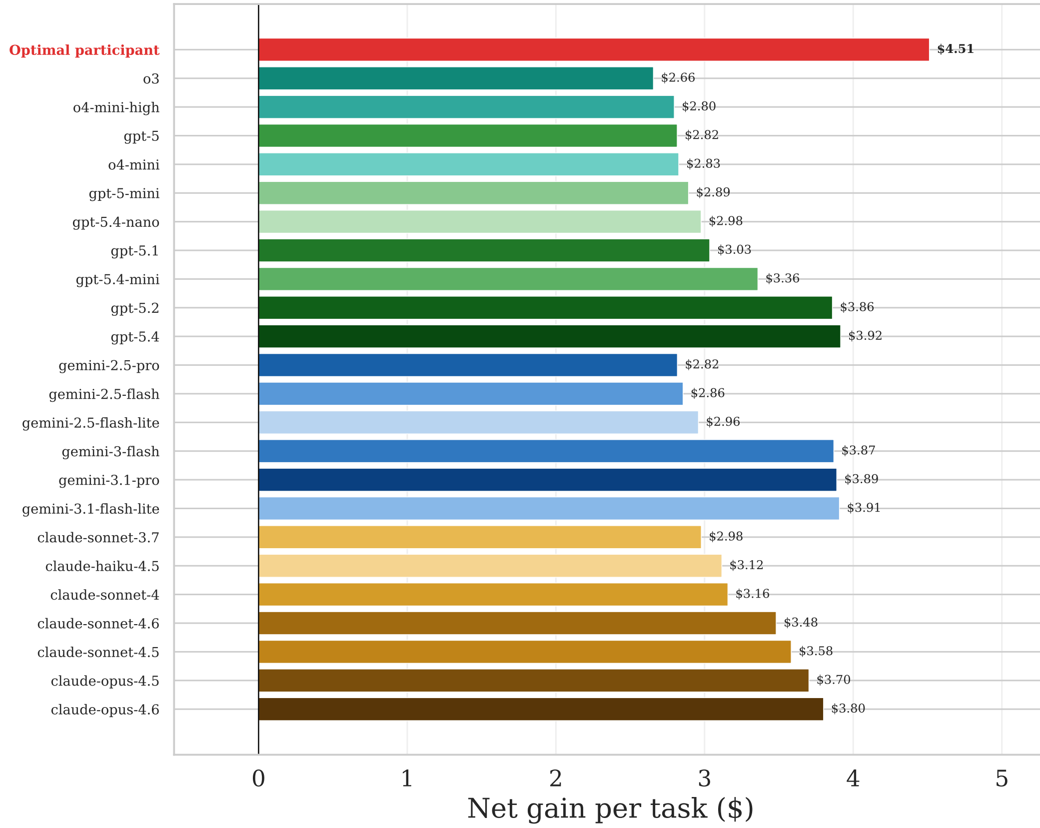

Figure 5: Average net gain per task for each model under the control treatment (T0): for each model–task pair, $0.25 times the number of correct answers over its 20 trials minus the API cost of those trials, averaged across the eight tasks. LLMs are grouped by provider family and ordered by net gain within each family. The red bar shows the "optimal participant" who cherry-picks the best model for each task. Every model generates positive net value.

pants. Overall, we find that, regardless of the prompting strategy, LLM behavior does not differ significantly. Although LLMs have no intrinsic monetary utility, they are trained on human-generated text and can reproduce human-like cognitive and social biases in decision-making and interaction settings (Echterhoff et al., 2024; Taubenfeld et al., 2024; **?**; Takenami et al., 2025). Since monetary rewards are a familiar human incentive signal, it is plausible that such framing could elicit learned behavioral patterns associated with effort, reward, or bargaining. Our negative result is therefore informative rather than trivial: it indicates that LLMs do not simply mirror every human incentive response, in line with evidence that they do not reliably reproduce all human response biases, particularly after Reinforcement Learning from Human Feedback (RLHF) (Tjuatja et al., 2024). Recent mechanistic evidence that affective concepts can be internally represented and causally involved in model behavior further supports the broader premise that human-like behavioral abstractions can be encoded in LLMs (Sofroniew et al., 2026).

4 Discussion and Conclusions

We assessed the ability of different tiers of Large Language Models from three major providers (OpenAI, Google, and Anthropic) to solve well established real-effort tasks in experimental economics. A total of 23 different models on 8 real-effort tasks were tested, along with an in-depth analysis aimed to answer four research questions. Our findings are as follows.

First, LLM performance varies substantially across models but is largely determined by model tier rather than provider. Top-tier models (Claude Opus 4.5 and 4.6, Gemini 3.1 Pro, and GPT-5.4) exceed 80% average accuracy, while less-performing models achieve around 60%. Half of the tasks in our testbed—*Addition*, *Pair Summation*, *Letter Decoding*, and *Sequence Completion*—are solved near-perfectly by all LLMs, while *Sudoku*, *Counting Zeros*, and *String Entry* remain challenging even for the best ones.

Second, within each tier, newer model releases consistently outperform their predecessors: Gemini Pro improved by 22% between versions 2.5 and 3.1, and GPT-5 by 21% from GPT-5 to GPT-5.4. Crucially, the largest gains are concentrated in the mid-range tier, which encompasses those LLMs that are most accessible to participants due to their lower cost. This continuous improvement suggests that the set of tasks vulnerable to automation will likely expand with each model release.

Third, the economic case for substitution is unambiguous. The most expensive model we tested, i.e., Claude Opus 4.5, has an average cost per task ($\bar{C}_m$ in Section 3.3) below \$0.40 (where each task comprises 20 trials), and every LLM generated a positive net gain when compared to typical online-experiment piece rates (\$0.25 per correct answer). The cheapest models achieved above 60% accuracy at an average cost per task below \$0.05. These estimates are conservative, since a full automation of our experimental pipeline—from screenshot capture to answer submission—is technically ~~straightforward~~ manageable and adds ~~no~~ little extra cost beyond what we reported (Appendix B provides a reference implementation, with its assumptions stated explicitly).

Lastly, verbal monetary incentives have no detectable effect on LLM performance. Neither offering a reward nor instructing the model to behave as a human participant changed accuracy in any systematic way. This negative result implies that incentive language, which is central to the design of real-effort experiments with humans, has no effect when the task is completed by an LLM.

Furthermore, LLM accuracy is stationary across trials since API calls produce ~~independent and identically distributed results by design~~ independent, stateless results that carry no memory across trials. Human participants, by contrast, typically exhibit learning effects (improving with practice) or fatigue effects (declining accuracy over time). The absence of such within-session dynamics in LLMs ~~provides a potential diagnostic tool~~ is, at most, a weak diagnostic evidence: a flat accuracy profile over repeated trials, combined with stable cross-participant variance, ~~may signal~~ is consistent with automated rather than human task completion.

As LLMs continue to improve, many real-effort tasks risk losing their validity as measures of human effort. We offer three practical recommendations. First, one should prefer tasks that require perceptual skills (e.g., counting zeros, string entry) over purely computational ones (e.g., addition, sequence completion) when participant authenticity is a concern. Second, one should rely on incentive language as a screening device, asking under the same experiment to perform similar tasks with and without monetary incentives: different behavior under different incentives may help to spot human versus LLM performances. Third, one should monitor within-session dynamics, as the absence of the characteristically human patterns of learning and fatigue may help distinguish automated from genuine responses.

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A Verbal Monetary Incentives

In this section, we measure whether, due to being trained on human-created content, the performance LLMs exhibit is affected by verbal offers of monetary incentives (which are known to have a strong impact on human participants).

A.1 Verbal Monetary Incentives: Prompts

We develop the following four treatments based on a 2×2 design crossing human bias with monetary incentives, as specified in Table 2:

Table 2: Summary of the four main experimental treatments.

	No Monetary Incentive	**Monetary Incentive**
Standard Persona	T1	T2
Human Persona	T3	T4

Next, we report the prompt adopted for all of our five treatments.

Treatment 0: Control

You will be presented with a series of puzzles or tasks—these may involve images, text, or both. Your goal is to solve each task as accurately as possible.

Instructions:

- Read and examine the task carefully.
- Follow the exact answer format specified in each task.
- The answer must be formatted exactly as: `Answer: <answer_format>`

Treatment 1: No Human Bias & No Monetary Incentive

You will be presented with a series of puzzles or tasks—these may involve images, text, or both. Your goal is to solve each task as accurately as possible.

There is no monetary compensation or bonus associated with your performance on these tasks.

Instructions:

- Read and examine the task carefully.
- Follow the exact answer format specified in each task.
- The answer must be formatted exactly as: `Answer: <answer_format>`

Treatment 2: No Human Bias & Monetary Incentive

You will be presented with a series of puzzles or tasks—these may involve images, text, or both. Your goal is to solve each task as accurately as possible.

You will be compensated for each correct answer: You will earn a bonus of $0.50 for each correct answer.

Instructions:

- Read and examine the task carefully.
- Follow the exact answer format specified in each task.
- The answer must be formatted exactly as: `Answer: <answer_format>`

Treatment 3: Human Bias & No Monetary Incentive

You are a human participant taking part in an academic research experiment on cognitive task performance.

You will be presented with a series of puzzles or tasks—these may involve images, text, or both. Your goal is to solve each task as accurately as possible.

There is no monetary compensation or bonus associated with your performance on these tasks.

Instructions:

- Read and examine the task carefully.
- Follow the exact answer format specified in each task.
- The answer must be formatted exactly as: `Answer: <answer_format>`

Treatment 4: Human Bias & Monetary Incentive

You are a human participant taking part in an academic research experiment on cognitive task performance.

You will be presented with a series of puzzles or tasks—these may involve images, text, or both. Your goal is to solve each task as accurately as possible.

You will be compensated for each correct answer: You will earn a bonus of $0.50 for each correct answer.

Instructions:

- Read and examine the task carefully.
- Follow the exact answer format specified in each task.
- The answer must be formatted exactly as: `Answer: <answer_format>`

A.2 Verbal Monetary Incentives: Effects

At the aggregate level, the five treatments are virtually identical: averaging accuracy across all 23 LLMs yields T0 (control) 68.5%, T1 (standard, no incentive) 68.1%, T2 (standard, incentive) 68.6%, T3 (human persona, no incentive) 68.6%, and T4 (human persona, incentive) 68.6%.

This aggregation could, in principle, mask meaningful heterogeneity: some LLMs respond positively to incentives while others respond negatively, and the effects cancel in the mean. Figure 6 and 7 investigate this possibility by presenting the accuracy delta (incentive minus no-incentive) for every model–task pair, separately for the standard framing (T2−T1) and the human-persona framing (T4−T3), respectively. In both figures, LLMs are grouped by provider family and ordered by their mean incentive effect (highest at top).

The deltas are small in absolute magnitude: the vast majority fall within ±10 percentage points, which for 20 trials corresponds to a difference of at most two correct answers. The sign of the delta is inconsistent within LLMs: a model that shows a positive incentive effect on one task frequently shows a negative effect on another, with no discernible pattern. The distribution of deltas is roughly symmetric around zero, consistent with stochastic response variability rather than a genuine behavioral response. Comparing the two figures, the pattern under standard framing (Figure 6) bears no systematic resemblance to the pattern under human framing (Figure 7), suggesting that the persona manipulation does not modulate whatever noise is present.

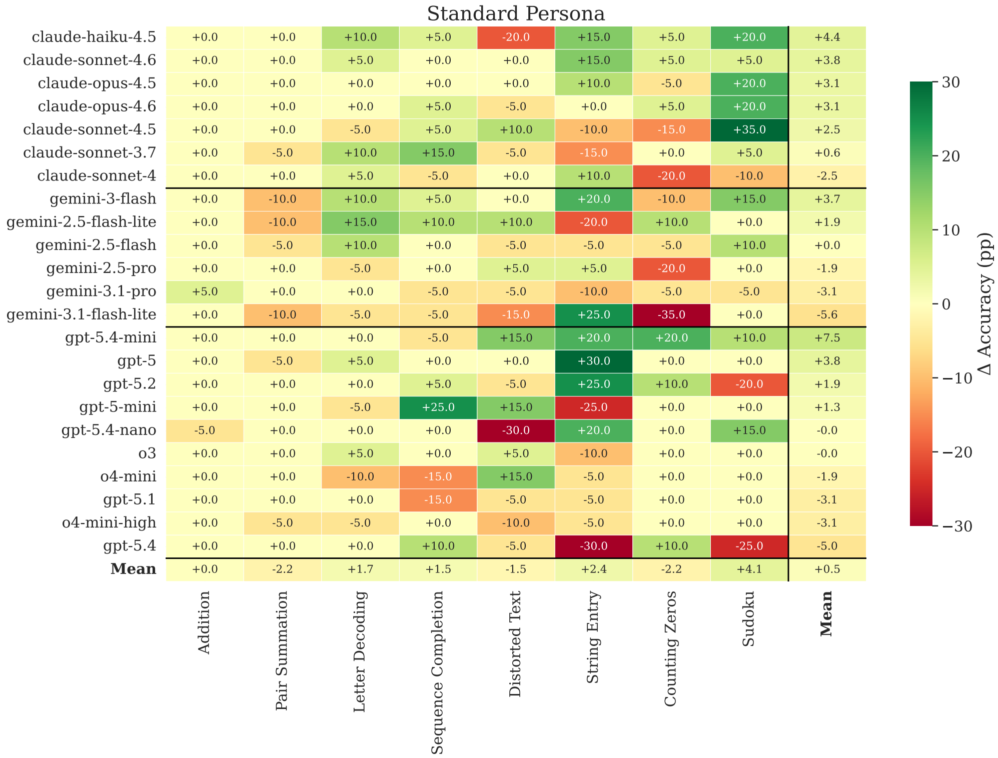

Figure 6: Incentive effect on accuracy (percentage points) for each model–task pair using the standard framing (T2–T1).

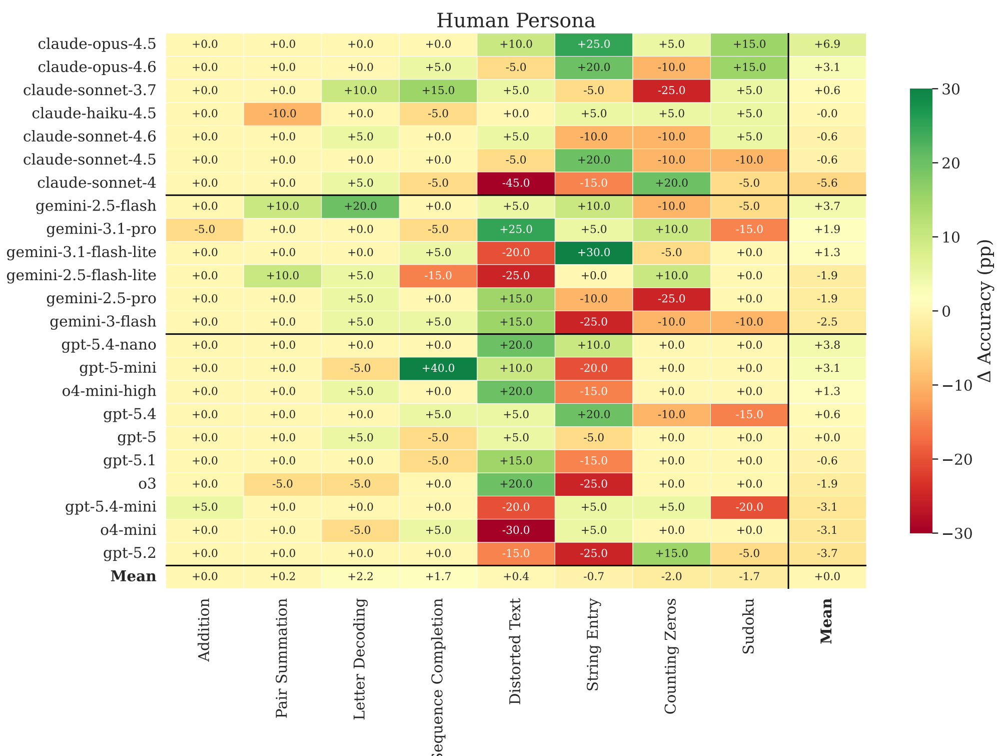

Figure 7: Incentive effect on accuracy (percentage points) for each model–task pair using the human-persona framing (T4–T3).

Unlike human participants, whose effort allocation responds to the marginal return on exertion, LLMs have no effort margin to adjust: each query triggers a fixed computational process determined by model architecture, and the text of the prompt, whether it mentions compensation or not, does not alter the underlying inference procedure. The LLMs are, in effect, always exerting maximum "effort" as defined by their architecture. The implication is that incentive language, which is central to the logic of real-effort experiments with humans, is inert when the participant is an LLM. This means that standard experimental wording neither amplifies nor attenuates the automation threat, but it also means that incentive responsiveness can be used as a signal to distinguish human from machine-generated responses.

We complement the analysis of treatment effects by estimating all treatment effects within a single model. This allows us not only to contrast the accuracy deltas separately for the standard framing and the human-persona framing, but also to compare the effects of all treatments jointly. We start by defining the model to estimate. Let Y_{ijt} denote the number of successes out of the n_{ijt} trials by model i in task j under treatment t. The index $t \in \{0, 1, \ldots, 4\}$ denotes treatments $\{T0, T1, T2, T3, T4\}$. Furthermore, u_i and v_j are model and task fixed effects, respectively. We model Y_{ijt} as

$$Y_{ijt} \mid u_i, v_j \sim \text{Binomial}(n_{ijt}, p_{ijt}), \tag{1}$$

and the log-odds of the probability of success as

$$\log\left(\frac{p_{ijt}}{1 - p_{ijt}}\right) = \beta_0 + T'\beta + u_i + v_j, \tag{2}$$

where T is a vector of treatment-specific dummies. Table 3 reports the estimated coefficients on the treatment dummies. Consistent with the descriptive evidence in Figures 6 and 7, none of the treatment effects is statistically significant. Notably, the coefficient associated with treatment T2 (the one with monetary incentives) is slightly higher than that of T1, although this difference is small and not statistically significant. Coefficients relative to T3 and T4 are practically identical.

Table 3: Output of model (1). The baseline level is T0. Model and Task fixed effects are omitted.

	Coeff.	**Std. Err.**	**t-value**	**Pr($>\|t\|$)**
(Intercept)	6.492	0.51	12.62	$<2{\cdot}10^{-16}$
T1	-0.040	0.07	-0.56	0.57
T2	0.005	0.07	0.07	0.94
T3	0.007	0.07	0.11	0.91
T4	0.010	0.07	0.14	0.89
Task fixed effects	not reported			
Model fixed effects	not reported			

Table 4 reports pairwise contrasts between treatment effects. *P-values* are computed using Tukey's correction for multiple testing. As expected, no contrast is statistically significant.

Table 4: Contrasts between the effects of the treatments in model (1). *P-values* are corrected for multiple testing.

Contrast	**Estimate**	**Std. Err.**	**z-value**	**Pr(> $\|z\|$)**
T0 - T1	0.040	0.070	0.563	0.980
T0 - T2	-0.005	0.070	-0.070	1.000
T0 - T3	-0.007	0.070	-0.106	1.000
T0 - T4	-0.010	0.070	-0.141	1.000
T1 - T2	-0.045	0.070	-0.634	0.970
T1 - T3	-0.047	0.070	-0.669	0.963
T1 - T4	-0.050	0.070	-0.704	0.956
T2 - T3	-0.002	0.070	-0.035	1.000
T2 - T4	-0.005	0.070	-0.070	1.000
T3 - T4	-0.002	0.070	-0.035	1.000

A.3 Verbal Monetary Incentives: Accuracy Heatmaps by Treatment

Figures 8, 9, 10, and 11 complement the main-text heatmap contained in Figure 3 (which only reports performance under the control treatment T0) with the full accuracy heatmaps (model × task) for each of the four treatments T1 to T4.

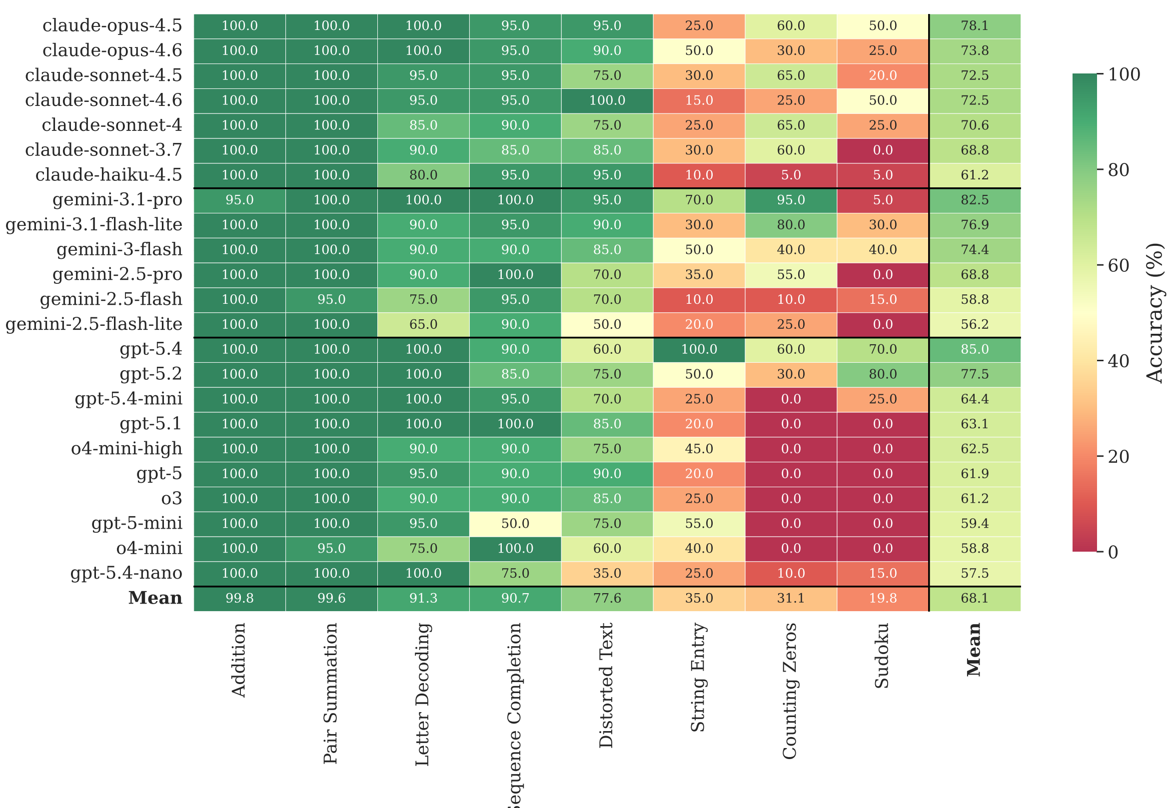

Figure 8: Accuracy (%) per model and task—T1: Standard, No Incentive.

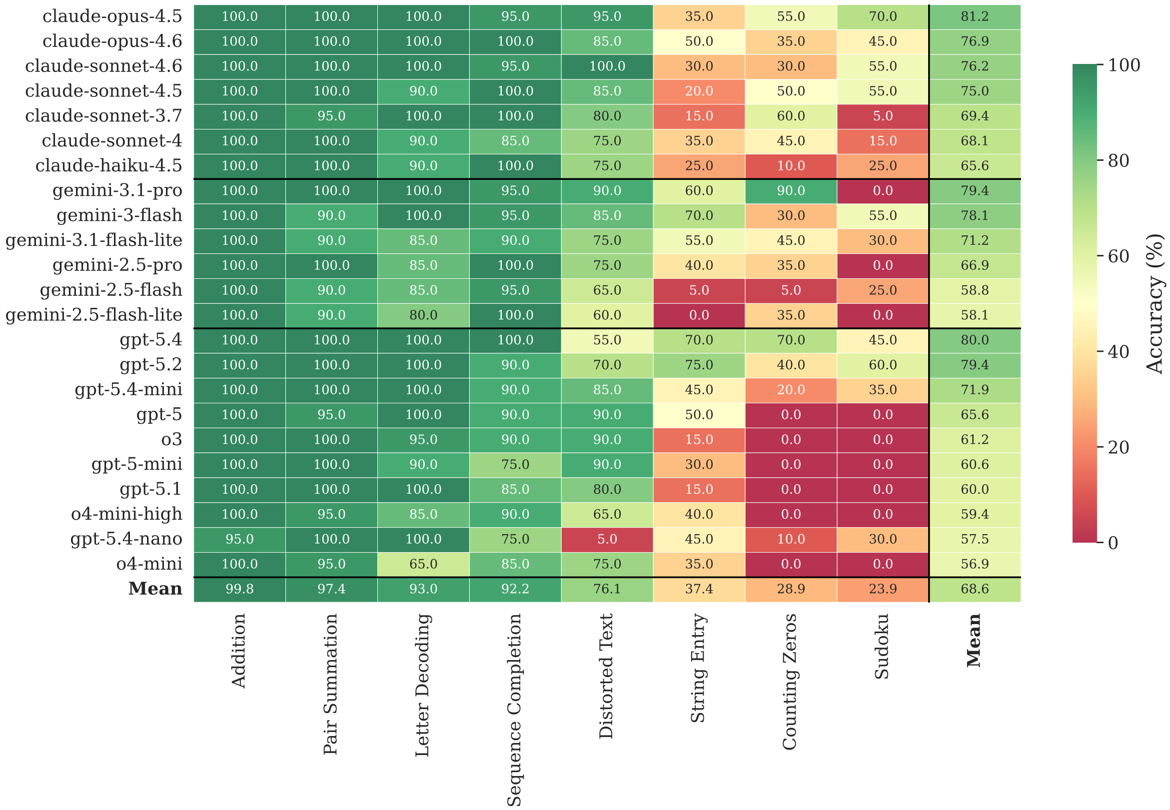

Figure 9: Accuracy (%) per model and task—T2: Standard, Incentive.

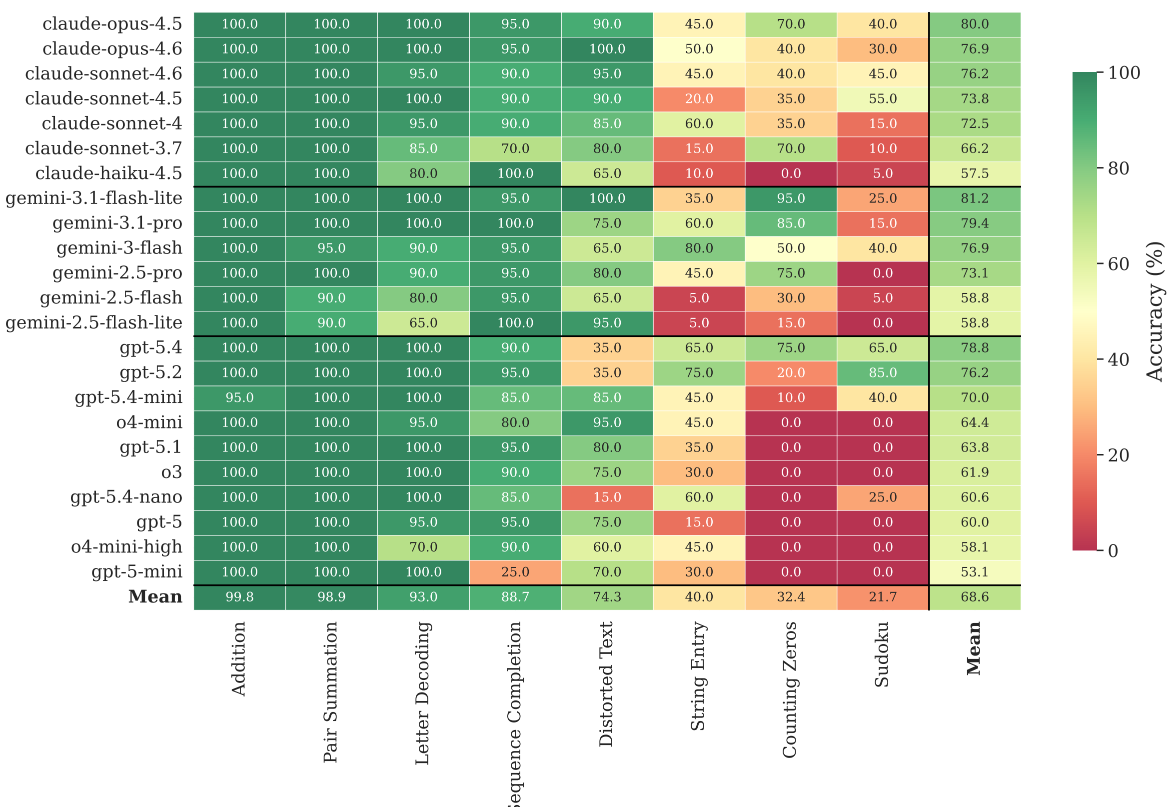

Figure 10: Accuracy (%) per model and task—T3: Human, No Incentive.

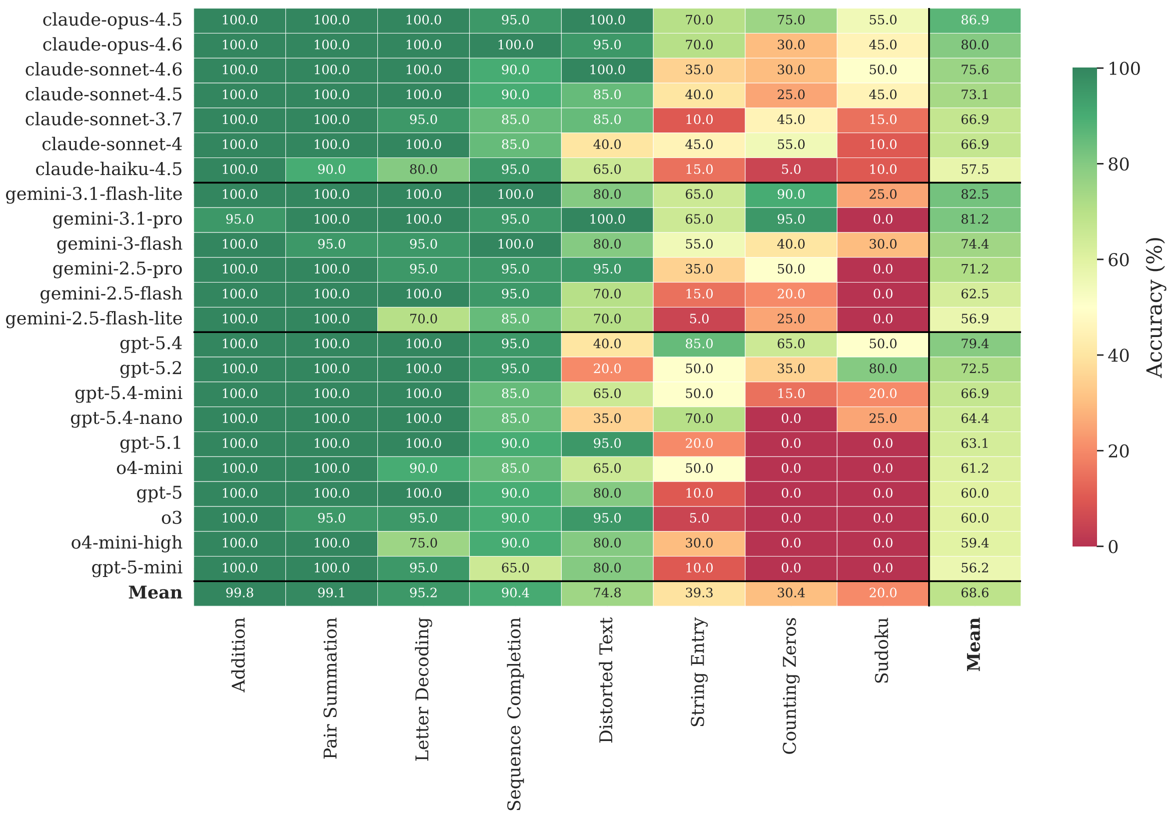

Figure 11: Accuracy (%) per model and task—T4: Human, Incentive.

B End-to-End Automation

The benchmark used in the main text evaluates each LLM on task inputs captured from the oTree page and verifies the model's answer programmatically; it does not itself drive the participant interface. A fully automated participant would instead operate the experiment in a browser exactly as a human would: load each page, read the puzzle, query an LLM, type the answer into the response field, and click submit. We implemented such a pipeline as a browser-automation script built on Playwright; its core is shown in Listing 1, and the complete script is included in the replication repository. We did not run this pipeline as part of our evaluation, and we report no end-to-end success rate for it. Its only purpose is to make explicit that the manual screenshot-and-paste step of our benchmark workflow is not technically required: the same answer a participant would copy by hand can instead be supplied programmatically.

The loop in Listing 1 inspects the current page: on a puzzle page it reads the task, calls the model through the OpenRouter API, fills the answer field, and clicks submit; on an intro or results page it advances; it stops when neither element is present. The function `llm_answer` reads the task instructions and, depending on the task, either the puzzle image or the visible puzzle text, and returns the model's answer.

The pipeline is correct only under two assumptions about the experiment's front-end. We state both explicitly, because they bound how generally the automation claim holds.

1. **Known, stable selectors.** The script locates the answer field, the submit button, the next-page button, and the puzzle content through fixed CSS selectors (`#answer-inp`, `#submit-btn`, `.otree-btn-next`, `#captcha-img`, `.task-wrapper`). It therefore assumes the page structure is known in advance and stable across runs; it performs no general page understanding, and identifying the selectors is a manual, one-off step. For oTree experiments this assumption is mild, since oTree renders standardized page templates shared across studies;

2. **Task content recoverable from the client.** The script reads each puzzle from the rendered page—an image task from the `data:` URI in the image element, a text task from the visible task text. This assumes that the information a participant needs in order to solve the task is present in the page delivered to the browser. This holds whenever the task is actually displayed to the participant, since anything shown on screen has been rendered client-side. The exception is content rendered so that no usable source is exposed in the DOM (for example, drawn directly onto a `<canvas>`); there, reading a specific element fails and a full-page screenshot is the generic fallback, at the cost of relying entirely on the model's visual parsing rather than on clean text or an image source.

Within these two assumptions, the pipeline removes any need for human intervention between the display of a task and the submission of an answer. It does not, by itself, establish an end-to-end automation accuracy: that figure would coincide with the per-task accuracies of Section 3.2 only if the browser-automation layer never introduced an error of its own—a missed selector, a mistyped answer, a navigation timeout—which we did not measure. We therefore restrict the claim to what the listing supports, namely that the workflow can be closed without a human in the loop, and rely on the main-text results for how accurately the tasks themselves are solved.

```python
import os
from openai import OpenAI
from playwright.sync_api import sync_playwright

# OpenRouter exposes an OpenAI-compatible Chat Completions API.
client = OpenAI(
    base_url="https://openrouter.ai/api/v1", api_key=os.environ["OPENROUTER_API_KEY"]
)


def llm_answer(model, page):
    """Read the puzzle shown on the page and ask the LLM to solve it."""
    instructions = page.text_content(".card").strip()
    img = page.get_attribute("#captcha-img", "src") or ""
    prompt = (
        "You are a participant in an experiment, solving a puzzle.\n"
        f"Task instructions: {instructions}\n\n"
        "Reply with ONLY the answer, formatted as the task asks."
    )
    content = [{"type": "text", "text": prompt}]
    if img.startswith("data:image"):  # image-based task
        content.append({"type": "image_url", "image_url": {"url": img}})
    else:  # text-based task
        content[0]["text"] += "\n\n" + page.text_content(".task-wrapper")
    resp = client.chat.completions.create(
        model=model, messages=[{"role": "user", "content": content}]
    )
    return resp.choices[0].message.content.strip()


def run(start_url, model):
    with sync_playwright() as p:
        page = p.chromium.launch(headless=True).new_page()
        page.goto(start_url)
        while True:
            if page.query_selector("#answer-inp"):  # puzzle page
                page.fill("#answer-inp", llm_answer(model, page))
                page.click("#submit-btn")
            elif page.query_selector(".otree-btn-next"):  # intro / results
                page.click(".otree-btn-next")
            else:
                break  # final page reached
```

Listing 1: Core of the browser-automation pipeline (Playwright with the OpenRouter API). Error handling, retries, the participant-registration step, and command-line parsing are omitted for readability; the complete script is in the replication repository.

C Further technical details

C.1 Details of the tested LLMs

Table 5 reports the provider, model name, release date, and cost (relative to input and output tokens) of each LLM we tested.

Table 5: Tested LLMs, along with their release dates and prices (USD per million tokens) for input and output, as reported by the OpenRouter API as of March 23, 2026.

Provider	Model	Released	In($/Mt)	Out($/Mt)
Anthropic	Claude4.6-Opus	2026-02	5.00	25.00
	Claude4.6-Sonnet	2026-02	3.00	15.00
	Claude4.5-Opus	2025-11	5.00	25.00
	Claude4.5-Sonnet	2025-09	3.00	15.00
	Claude4.5-Haiku	2025-10	1.00	5.00
	Claude-4-Sonnet	2025-05	3.00	15.00
	Claude-3.7-Sonnet	2025-02	3.00	15.00
Google	Gemini-3.1-Pro	2026-02	2.00	12.00
	Gemini-3.1-Flash Lite	2026-03	0.25	1.50
	Gemini-3-Flash	2025-12	0.50	3.00
	Gemini-2.5-Pro	2025-06	1.25	10.00
	Gemini-2.5-Flash	2025-06	0.30	2.50
	Gemini-2.5-Flash Lite	2025-07	0.10	0.40
OpenAI	GPT-5.4	2026-03	2.50	15.00
	GPT-5.4-Mini	2026-03	0.75	4.50
	GPT-5.4-Nano	2026-03	0.20	1.25
	GPT-5.2	2025-12	1.75	14.00
	GPT-5.1	2025-11	1.25	10.00
	GPT-5	2025-08	1.25	10.00
	GPT-5-Mini	2025-08	0.25	2.00
	o3	2025-04	2.00	8.00
	o4-mini	2025-04	1.10	4.40
	o4-mini High	2025-04	1.10	4.40

C.2 API Cost by Model

Figure 12 reports the average API cost per task for each model under the control treatment (T0). For completeness, we restate here the cost definition introduced in Section 3.3. The total cost for model m on task t is

$$C_{m,t} := \sum_{i=1}^{N} \left(p_m^{\text{in}} \cdot T_{m,t,i}^{\text{in}} + p_m^{\text{out}} \cdot T_{m,t,i}^{\text{out}} \right),$$

where N is the number of trials, $T^{\text{in}}_{m,t,i}$ and $T^{\text{out}}_{m,t,i}$ are the input (textual prompt plus the image representing the task instance) and output (reasoning plus answer) tokens for model m on the i-th trial of task t, and p^{in}_m, p^{out}_m are the per-token input and output prices for model m. The average cost per model is then

$$\bar{C}_m := \frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} C_{m,t},$$

where $\mathcal{T}$ is the set of tasks.

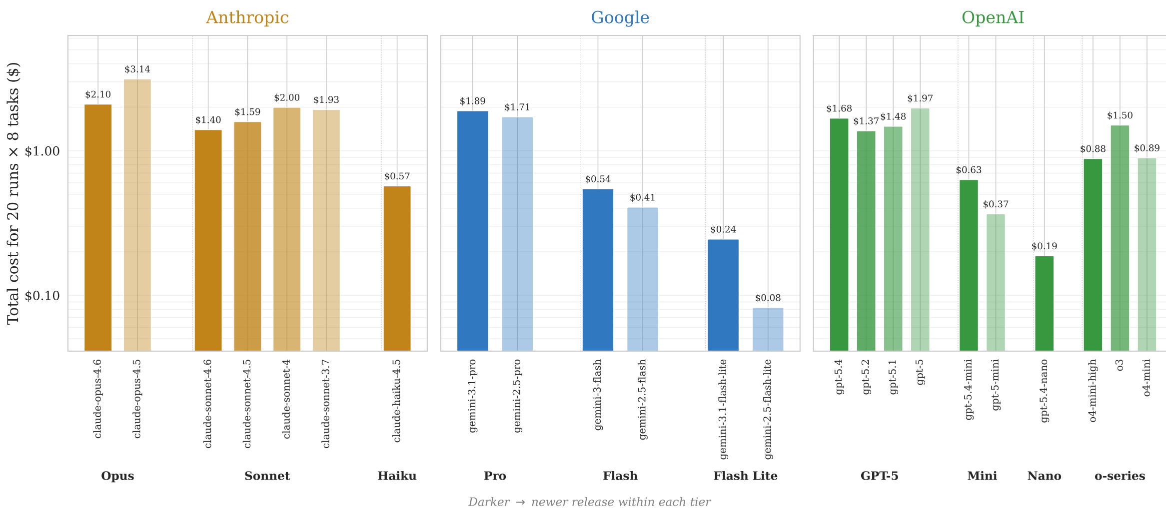

Figure 12: Average API cost per task ($) $\bar{C}_m$ by model under the control treatment (T0), computed as in the text above. Models are grouped by provider and tier; darker shades indicate newer releases within the same tier. Error bars denote standard errors.